

● EASY

● INFORMATION AND IDEAS

Information and Ideas

09

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

Many trees produce growth rings as they age, with each ring in a tree's trunk representing one year in the tree's life. This often makes it fairly easy to determine how old a tree was when it was cut down. To do so, you look at the tree stump and count the dark rings you see. But a researcher claims that this method often can't be used to identify the age of olive trees.

Which detail, if true, would most directly support the researcher's claim?

- A. The oldest olive tree in the world is likely over 1,100 years old.
- B. Narrow growth rings can suggest that an olive tree experienced harsh conditions.
- C. Many olive trees have growth rings that are difficult to see.
- D. Olive trees thrive in areas with hot, dry summers.

Hevea brasiliensis, a tree in the Amazon rainforest, is the world's main source of natural rubber. The tree produces a milky substance called latex that is used to make rubber. The bark of *Hevea brasiliensis* is helpful for the process of making rubber because it has a unique structure that makes it easy to collect latex. A network of tubes in the tree's inner bark helps the latex to flow out easily when people make small cuts into the bark.

What feature of *Hevea brasiliensis* does the text say is helpful for the process of making rubber?

- A. Its latex produces rubber of an especially high quality.
- B. Its bark has a unique structure that makes it easy to collect latex.
- C. It is able to grow in a wide variety of climates around the world.
- D. It is one of only two trees in the Amazon that produce latex.

Number and Origin of Clamshell Tools Found at Different Depths
below the Surface in Neanderthal Cave

Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
2–3	7	0
3–4	99	33
4–5	2	0
5–6	18	7
6–7	1	0

Two kinds of clamshell tools used by Neanderthals were dug up in a cave on the western coast of Italy. Archaeologist Paola Villa and her colleagues studied the tools and determined that Neanderthals either collected clams that had washed onto the beach or harvested clams from the seafloor and then sharpened the shells to make tools. The highest number of tools made from clamshells that were collected from the beach was found at a depth of _____blank

Which choice most effectively uses data from the table to complete the text?

- A. 5–6 meters below the surface.
- B. 4–5 meters below the surface.
- C. 3–4 meters below the surface.

D. 6–7 meters below the surface.

In the 1700s and 1800s, European composers experimented with volume in their musical works. They did so by increasing the number of musicians playing in the orchestra. For example, in some of his operas, German composer Richard Wagner added more horns, trombones, and tubas to the orchestra. With more instruments playing at the same time, the orchestra could play extremely loudly at key moments in his operas.

According to the text, how did Richard Wagner achieve moments of extremely high volume in his operas?

- A. By moving the performances of his operas from outdoor stages to indoor ones
- B. By increasing the number of musicians playing horns, trombones, and tubas in the orchestra
- C. By building a concert hall whose shape would cause sounds to echo
- D. By insisting that the singers undergo special training to sing for extended periods of time

Olms are salamanders that live in underwater caves. Scientists once thought that olms stay in their caves all their lives. However, Raoul Manenti and team claim that olms regularly come to the surface to perform important activities such as finding food.

Which finding, if true, would most strongly support the underlined claim?

- A. Researchers determine that olms don't breed often.
- B. Researchers confirm that olms live in only a few cave systems.
- C. Researchers discover that earthworms from surface soils are a major part of olms' diet.
- D. Researchers learn that olms' brains differ from other salamanders' brains.

Scrapbooks of saved fabric pieces were commonly kept by women in the nineteenth-century United States, but few are as meticulously detailed as Hannah Ditzler Alspaugh's work. Alongside each piece of fabric, Alspaugh recorded intimate memories, such as dressmaking with her sister. Additionally, she listed the prices and how she used the fabric. Historians note that by representing fifty years of changing textures, patterns, and dress styles, the scrapbook is a record of nineteenth-century textiles and dressmaking as well as Alspaugh's life.

Which choice best states the main idea of the text?

- A. Alspaugh inspired other women to save pieces of fabric in scrapbooks and provide historical records of nineteenth-century fashions in the United States.
- B. Historians rely on fabric scrapbooks to understand how fashions changed throughout the nineteenth-century United States.
- C. Fabric scrapbooks were a popular hobby for many women in the nineteenth-century United States.
- D. Alspaugh's scrapbook provides a detailed account of her life and historical record of fashion trends in the nineteenth-century United States.

Total Areas of Five Tribal Nations in California

Tribal nation	Location	Area (square miles)
Hoopa Valley Tribe	Northern California	141.68
La Jolla Band of Luiseño Indians	Southern California	13.50
Pauma Band of Luiseño Mission Indians	Southern California	9.36
Agua Caliente Band of Cahuilla Indians	Northern California	53.68
Los Coyotes Band of Cahuilla and Cupeño Indians	Southern California	39.21

In what is now the state of California, there are over 100 tribal nations. One of the largest, the Hoopa Valley Tribe, covers 141.68 square miles in the northern part of the state. However, most tribal nations in California are less than 50 square miles in total area. For example, the total area of the Los Coyotes Band of Cahuilla and Cupeño Indians in Southern California is _____blank

Which choice most effectively uses data from the table to support the statement?

- A. 13.50 square miles.
- B. 53.68 square miles.
- C. 9.36 square miles.

D. 39.21 square miles.

The following text is adapted from Frances Hodgson Burnett's 1911 novel *The Secret Garden*. Mary, a young girl, recently found an overgrown hidden garden.

Mary was an odd, determined little person, and now she had something interesting to be determined about, she was very much absorbed, indeed. She worked and dug and pulled up weeds steadily, only becoming more pleased with her work every hour instead of tiring of it. It seemed to her like a fascinating sort of play.

Which choice best states the main idea of the text?

- A. Mary hides in the garden to avoid doing her chores.
- B. Mary is getting bored with pulling up so many weeds in the garden.
- C. Mary is clearing out the garden to create a space to play.
- D. Mary feels very satisfied when she's taking care of the garden.

Percent of Residents of City Areas in Favor of Adding More Bike Paths

City Area	Percent of area's residents in favor of adding more bike paths
North East	12%
North Central	26%
North West	46%
South West	88%
South Central	33%

A city's Parks and Recreation department is interested in providing residents with more opportunities for bicycling in their neighborhoods. They're considering adding more bike paths and conducted a survey to understand where demand for more bike paths is highest. The survey indicated the highest level of demand, with 88 percent of the residents interested in adding more bike paths, is in the city's _____ blank

Which choice most effectively uses data from the table to complete the statement?

- A. South West area.
- B. South Central area.
- C. North East area.
- D. North Central area.

NASA's *Cassini* probe has detected an unusual wobble in the rotation of Mimas, Saturn's smallest moon. Using a computer model to study Mimas's gravitational interactions with Saturn and tidal forces, geophysicist Alyssa Rhoden and colleagues have proposed that this wobble could be due to a liquid ocean moving beneath the moon's icy surface. The researchers believe other moons should be examined to see if they too might have oceans hidden beneath their surfaces.

Which choice best states the main idea of the text?

- A. Rhoden and colleagues were the first to confirm that several of Saturn's moons contain hidden oceans.
- B. Research has failed to identify signs that there is an ocean hidden beneath the surface of Mimas.
- C. Rhoden and colleagues created a new computer model that identifies moons with hidden oceans without needing to analyze the moons' rotation.
- D. Research has revealed that an oddity in the rotation of Mimas could be explained by an ocean hidden beneath its surface.